IMA EDUCATIONAL Case Journal | Case Study

IMA INSTITUTE OF MANAGEMENT ACCOUNTANTS Advancing the Profession™

ISSN 1940-204X

Aero Gear, Inc.: Performance Measurement, Cost Management and Product Costing in a Lean Transition

Larry Grasso
Central Connecticut State University

"Our labor efficiency numbers have lost their meaning. We just don't know what to replace them with."
Doug Rose, President of Aero Gear.

In January 2001, Aero Gear's President Doug Rose reviewed the monthly and year-to-date financial statements and performance reports with John McDermott, Vice President of Sales and Marketing, and Everett Smith, Vice President of Operations. Doug and his management team had become increasingly dissatisfied with the performance reports. In the past, they felt the reports had provided reliable information on labor efficiency and on the profitability of their jobs. In 1998, however, Aero Gear had begun a transition to Lean business practices. The accounting system and reports had not changed. Doug, John and Everett did not trust the numbers reported by their accounting system in the new production environment. They felt they had lost the ability to monitor the efficiency and productivity of their machinists and they were no longer confident they knew which jobs were most profitable and which were unprofitable. They wanted their accounting system to provide a means to monitor productivity and efficiency and manage costs, and to provide accurate product costs.

BACKGROUND

With the encouragement of Pratt & Whitney, which was looking for a local source for carburized and hardened gears, Doug Rose and his father Walter founded Aero Gear in 1982. In 1995, Aero Gear expanded into gearbox assembly but the company continued to be primarily a manufacturer of precision gears and gear shafts. Examples of the parts Aero Gear produces are pictured in Exhibit 1.

Exhibit 1

Sample Products



Gear box Spiral bevel gear Spur gears



Spur gears Spiral bevel gear Couplings



Splined parts Pump gears Gears and gear shafts

Competitive pressure on Aero Gear rose during the 1990s. Aero Gear's customers were engaged in fierce competition for market share in the commercial aerospace sector. They, in turn pressed for cost reductions from their suppliers. In addition, foreign commercial and defense aerospace customers increasingly insisted on local sourcing of significant amounts of work as a precondition for accepting bids from aircraft and engine manufacturers. Diverting subcontract work away from their U. S. suppliers helped satisfy these demands. The fact that Aero Gear was located in a high wage state added to the competitive pressure.

In response Doug Rose spearheaded the development of the Aerospace Components Manufacturers organization in 2000. Aero Gear and other local aerospace suppliers joined together to improve overall efficiency and to work cooperatively to attract business beyond the capabilities of any one member company. The ACM members decided that adopting and implementing lean business practices would be the foundation of their efforts to jointly improve their competitive position.

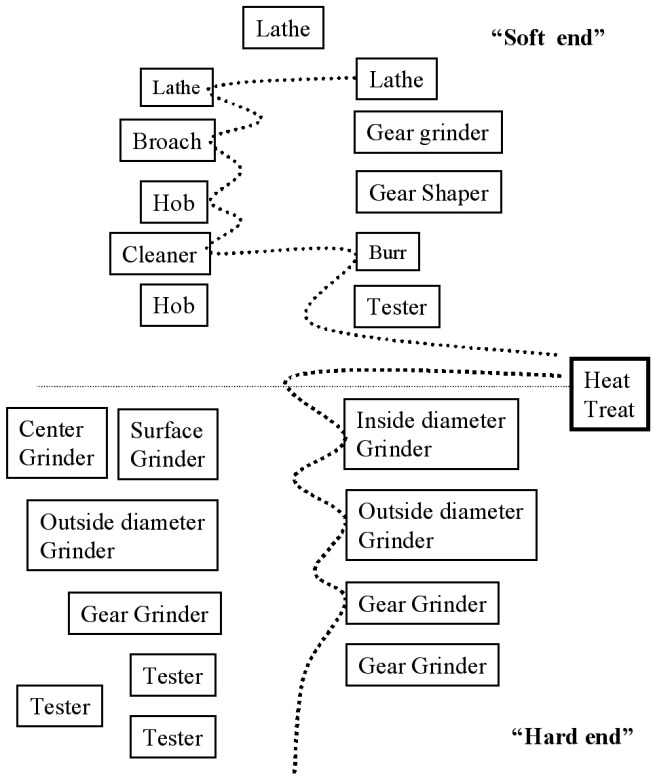
Aero Gear's 90 employees generated $16 million sales during 2001 from its modern 52,000 square foot facility located in a Windsor, Connecticut industrial park. Two thirds of their sales were for aviation applications (43% commercial, 24% military). The remaining third of their sales were for land-based turbine engines (28%) and rockets (5%). Pratt & Whitney remained an important customer along with sister United Technologies Corporation divisions Hamilton Sundstrand and Sikorsky Aircraft, but Aero Gear also supplied other customers including Raytheon, Boeing, and European aerospace companies. They produced approximately 175 different part numbers during 2001.

THE TRANSITION TO LEAN MANUFACTURING

Prior to beginning their transition to lean business practices, Aero Gear had been organized as a job shop. Machines were clustered by type. A machinist would perform an operation for a job at one machine. When that operation was completed, the partially completed parts would be moved to a queue near a machine operated by another machinist that would perform the next operation, and so on. Like any typical batch and queue environment, Aero Gear's shop had large amounts of work-in-process, long wait times and lead times, and jobs traveling long distances back and forth across the shop before they were completed.

In March 1998, Aero Gear began their transition to lean. They identified three value streams, and production cells were created for each. Two value streams, "Bohica" and "Blue," covered a majority of Aero Gear's gear and gear shaft production. Due to the nature of their machinery and processes, the Bohica and Blue cells were organized as "flow lines" (machinists working in an aisle with machines on both sides) rather than the preferred U-shaped configuration. The Bohica value stream produced relatively simple parts, such as spur gears, pump gears, and couplings. The Blue value stream produced Spiral bevel gears and other complex gears requiring more machining operations. The flow lines were split into "soft" and "hard" ends, which were viewed as separate production cells within the value streams. After production was completed in the soft end, the parts were heat treated for hardening. The parts returned from heat treatment to the hard end of the line for final processing. The Bohica flow line configuration is presented in Exhibit 2 as an illustration.

Exhibit 2
The Bohica Flow Line
(Showing the path of a representative product)



Gearbox Assembly was identified as a third value stream, and a production cell was created for gearbox assembly. Machines not moved to the Bohica and Blue flow lines or the Assembly cell remained in a job shop setting. The job shop area produced gearbox housings, tooling, and parts not suited to either the Bohica or Blue value streams. About twenty

percent of Aero Gear's production remained in the job shop area during 2001.

Aero Gear's managers hired lean consultants to provide training and lead initial kaizen blitz events. All employees had gone through basic training in lean principles. Aero Gear was averaging about four kaizen blitzes per year in each value stream, most focused on achieving flow, reducing set-up times and reducing cycle times. In addition to the kaizen blitzes, employees implemented many small improvements related to flow, cycle times, setup times, mistake-proofing and other quality improvements. Setup times and cycle times were reduced, but great opportunities for further set-up time reduction remained. Employees in the cells were cross-trained (skills matrices were developed for employees in each cell), and 5S workplace reorganization and total productive maintenance were implemented. Employees were also beginning to develop and document standardized work, and were taking responsibility for quality, self inspecting after each process before parts were moved to the next process. Piece counts and quality metrics were tracked in the cells. Process improvements within the cells reduced non-value-added work and improved quality.

Because of machine constraints and the diversity of the parts produced by Aero Gear, considerable heterogeneity remained in the Blue and Bohica product families that impeded work flow. For example, a part otherwise suited for production on the Bohica line might have to go to the Blue line or the job shop for one operation, because the only machine capable of performing the operation was located there. In other cases there was "back flow" where a part had to return to the soft end of the flow line for an operation after heat treatment and some processing in the hard end of the line. Parts requiring outside processing (mostly plating) also disrupted production flow. Aero Gear was also confronted with changes in delivery requests from customers on short notice. Aero Gear's customers (often reacting to requests from their airline and government customers) would on short notice request delays or cancellation on orders for some parts and rapid acceleration of the delivery schedule on others.

Aero Gear had not established a pull system for materials and parts. Aero Gear was a small customer with little leverage to negotiate frequent shipments, and vendors for major materials had lead times as long as 50 weeks. They established FIFO lanes in front of both the soft and hard cells of the flow lines. They had not established a level of standard work-in-process for the flow lines in part because units flowing through each line varied significantly in cycle time.

Despite the challenges Aero Gear encountered, by 2001 the transition to lean was yielding improved operating results, especially in the Bohica value stream, where there were fewer problems with flow. Lead times for many parts produced in the Bohica value stream had been reduced from twelve weeks to four weeks, with a corresponding reduction in WIP inventory.

MANAGING PRODUCTION EFFICIENCY

Aero Gear's management relied primarily on labor reports to manage production efficiency. Production employees would complete time reports by logging on to terminals located on the shop floor. They would enter a part, job, and run number for each part they had been working on and an operation number for each operation they had performed. They also had ten time codes used to identify time spent on setup, making tooling, or to indicate abnormal conditions (for example, RW was used to identify time spent on rework, RP was used to identify time spent on a production run problem). Sixteen additional time codes were used to identify time spent on various indirect production activities (for example, machine maintenance, meetings, and sweeping/cleaning). Employees had to apportion all of the time they worked to direct labor operations and the indirect labor categories. The employees typically logged on once or twice per day to complete these reports.

The time entry data were used to generate two reports to monitor production efficiency: the Department Performance Report and the Summary Hours Report. The Department Performance Report was a detailed list of all direct labor activity with summary totals by employee, shift, and department. The report was generated daily and weekly. The report format is illustrated in Exhibit 3. Two particularly notable items on the report are the Direct Labor (D.L.) Factor and the Efficiency Percentage.

The D.L. Factor was computed as shown at the bottom of Exhibit 3. In the numerator, the cost of materials and any outside tooling and processing are deducted from the quoted price for each job. Thus, the measure implies Aero Gear charges customers for materials and outside processing at cost, and all profits earned by Aero Gear are the result of value added through internal processing and assembly. The D.L. Factor numerator was the measure of throughput or "value added" – what the customer was willing to pay above the cost of materials and outside processing. The D.L. Factor denominator was the budgeted direct labor hours for the job multiplied by $55, the historical conversion cost rate. Thus the D.L. Factor denominator is a measure of Aero Gear's cost of adding value through internal processing.

Exhibit 3

AERO GEAR INC.

Department Performance Report

Mm/dd/year – Mm/dd/year

Emp.#	Name	Date	Shift	Code	Dept.	AG#	Job#	Run#	Operation	Pieces	Hours	Actual Pcs/Hr	Est. Pcs/Hr	D.L. Factor	Budget Rate	Actual Rate	Dollars Earned	Efficiency Percentage
1234	Smith	mdy	2		24	2271	121	22	120	10	5.10	1.96	1.41	1.38	75.90	105.55	538.31	139.1%
1234	Smith	mdy	2	SU	24	2271	121	22	120	0	5.20		4.46	1.38	75.90	65.12	338.62	85.8%
1234	Smith	mdy	2		24	2271	121	22	120	15	8.60	1.74	1.41	1.38	75.90	93.89	807.45	123.7%
* Employee totals											18.90					84.64	1,684.38	117.4%
** Shift totals											184.00					80.91	14,886.94	88.0%
*** Department totals											539.40					84.28	45,461.46	90.4%

$$\textbf{D.L. Factor} = \frac{\textbf{Sales Price – Material Cost – Outside Tooling – Outside Processing}}{\textbf{(Estimated direct labor hours*) x 55}}$$

*(Inside tooling hours + Setup hours + Processing hours)

For example, suppose a job had the following information:

Selling price	$11,080
Material cost	$ 4,000
Outside tooling	$ 800
Outside processing	$ 1,000
Estimated inside tooling hours	16
Estimated setup hours	4
Estimated processing hours	40

The D.L. factor for the job would be 1.6, as shown below

$$\textbf{D.L. Factor} = \frac{\textbf{11,080 – 4,000 – 800 – 1,000}}{\textbf{(16 + 4 + 40) x 55}}$$

The **budgeted rate** for dollars earned per direct labor hour on the job would be $88.00 (11,080 – 4,000 – 800 – 1,000/(16 + 4 + 40), or $55 times the D.L. factor.

A worker completing a setup for this job in 24 minutes that was estimated to take 30 minutes would have an **efficiency percentage** of 125%, and an **actual rate** for dollars earned of $110.00 (125% of $88.00) per hour.

A worker completing 9 parts per hour for this job in a process that was estimated to complete 10 parts per hour would have an **efficiency percentage** of 90%, and an **actual rate** for dollars earned of $79.20 (90% of $88.00) per hour.

Budgeted labor hours included hours for setup and producing tooling in house as well as the hours for performing each production operation. For new parts, the budgeted hours were the estimated hours from the bid or job quote. For reorders of existing parts, the budgeted hours were based on past experience. The average hours for the three most recent production runs were used for parts that had been produced three or more times. The $55 conversion rate had not changed in about three years.

John McDermott commented on the D.L. Factor.

"We knew our current conversion rate was higher than $55, but we could adjust for that and the D.L. Factor still provided a good index for job profitability. The actual rate is closer to $75. Jobs with a D.L. Factor greater than 1.8 or so are our big profit makers, while a job with a D.L. Factor of 1.4 is pretty much breaking even. Anything below 1.4 is a loser. That knowledge is helpful for negotiating prices the next time that part is ordered and when preparing quotes for similar parts."

The Efficiency Percentage was computed for each operation by dividing the actual pieces per hour by the estimated pieces per hour.[1] Producing exactly to budget resulted in a 100% efficiency rate. Beating the budgeted time yielded an efficiency rate over 100%, and exceeding the budgeted time yielded an efficiency rate below 100%. The Efficiency Percentages were used to identify problem operations on a given part because the Department Performance Report listed Efficiency Percentages for each operation performed by each employee. The report was also used to identify particularly efficient or inefficient employees by looking for employees with consistently high or consistently low efficiency ratings or by comparing efficiency ratings for different employees performing the same operation. John McDermott commented:

"The Efficiency Percentage combined with the D.L. Factor quickly told us how we're doing on each run of a particular part. A part with a D.L. Factor of 1.8 should be profitable, but if we're only getting 50% Efficiency Percentages, we're losing money on this run."

A "Dollars Earned" metric also appeared on the department performance report. The total "value added" (or throughput, sales minus material and outside processing costs, the D.L. rate numerator) for each job was divided by the estimated direct labor hours required to complete the job, yielding a "Budget Rate" for work performed on the job. Employees were assigned dollars earned at the budget rate times the budgeted hours required to complete the operation. The "Actual Rate" is the budgeted rate multiplied by the efficiency percentage. Thus, the actual rate is a measure of value added dollars earned per actual hour worked performing a given operation.

The Summary Hours Report provided classified data on all hours worked by all employees. The report was generated weekly, showing the results of the most recent week along with the preceding eleven weeks and a total for the twelve-week period; and monthly, with totals for the most recent twelve-month period. The report format is illustrated in Exhibit 4. The total hours worked by each employee for each period were classified as direct (other than setup), setup or indirect. The average Efficiency Percentage for the direct and setup hours worked is also reported. Summary totals are presented by department and for the entire company. Doug Rose commented on the use of the Summary Hours Report.

"We used the Summary Hours Report to evaluate production employees because it displayed average efficiency rates for each employee and the trend in these rates over time. We also looked for departmental and company-wide improvements in the proportion of direct hours compared to setup hours and indirect hours. Setup and indirect activities are non value-adding activities, so we wanted to see those activities minimized."

LABOR REPORTING UNDER LEAN PRODUCTION

The lean production environment provided great opportunities for improvement, but it led to problems in labor reporting. Doug Rose discussed the reporting problems.

"In the job shop setting, we had one man, one machine reporting. Employees were doing one job at a time, and the times they reported were pretty reliable. Now, a machinist may be monitoring runs on two machines while setting up a third. When he completes his time report, he has to divide his time among those three operations. The computer will suggest one third of the time be allocated to each operation, but that may not be accurate, and the employee can override the default allocation. How does he decide to make that allocation? Well one thing he's likely to do is check what the budgeted times are for the operations and make sure if he can, that he's at or under budget for each operation. Now the times reported for all three operations are suspect."

John McDermott discussed how the labor reporting affected product cost and lead-time measures.

[1] In the case of setups or tooling, the Efficiency Percentage was computed as estimated time divided by actual time.

Exhibit 4 - AERO GEAR, INC. SUMMARY HOURS REPORT

		Week 1	Week 2	Week 3	... Week 12	
Area Employee	Type	mdy - mdy	mdy - mdy	mdy - mdy	mdy - mdy	Total
Blue Jones	Dir. Labor	45	50	57	33	429
1235	Setup	9 (20%)	10 (20%)	12 (20%)	9 (27%)	97 (23%)
1235	Indirect	12 (21%)	12 (20%)	6 (10%)	20 (38%)	221 (34%)
1235	Total	56	61	63	53	650
1235	Eff. %	50.20%	124.60%	143.60%	101.50%	95.10%
Area Direct hours		719	705	727	571	7,472
Area Setup hours		185 (26%)	119 (17%)	117 (16%)	118 (21%)	1,673 (22%)
Area Indirect hours		236 (25%)	269 (28%)	258 (26%)	332 (37%)	3,908 (34%)
Area Total hours		956	974	985	902	11,381
Area Efficiency %		68.60%	86.40%	94.90%	141.40%	91.80%
Company Direct hours		1,707	1,657	1,775	1,282	17,915
Company Setup hours		326 (19%)	264 (16%)	281 (16%)	225 (18%)	3,250
Company Indirect hours		1,103 (39%)	1,136 (41%)	1,106 (38%)	1,390 (52%)	3,908 (18%)
Company Total hours		2,809	2,794	2,881	2,672	15,207 (46%)
Company Efficiency %		79.90%	96.50%	103.10%	142.90%	89.20%

"Now we can't be sure that the times reported for an operation really reflect how long it takes. Suppose a machinist spends four hours monitoring runs on two machines. Two hours of labor time is assigned to each operation, but each operation actually takes four hours. It's the labor time that gets reported, and it's the labor time that gets used in revising budgeted times. The lead-time calculations will be off and the D.L. Factor loses its meaning. The time and cost allocated to one job now depends on what other jobs happened to be running in the cell at the same time. What happens if it runs alone or with a different job next time?"

Because the budgeted times were now suspect, Everett Smith undertook the task of reviewing all active parts to make sure the setup times and process times for each operation reflected the actual process time rather than a share of reported labor hours.

Doug Rose was also unhappy with the summary hours report.

"If our measuring system had any value you'd expect to see a change in efficiency, or in the proportion of hours spent on setups or indirect activities, but I don't see any significant change over the past year.

We need a better way to measure what it is actually costing to produce each job so that we'll know which jobs are generating profits and we'll be able to assess the performance of our flow lines [value streams]. We've started to produce P&L statements for the flow lines and the job shop each month, but I don't trust the numbers. We need to do a better job allocating costs to the lines and to the parts they produce. We also need better measures to evaluate the productivity and efficiency of the flow lines. Our labor efficiency numbers have lost their meaning. We just don't know what to replace them with."

PERFORMANCE MEASUREMENT AND OPERATIONAL CONTROL ON THE FLOW LINES

Aero Gear's managers were interested in finding more measures to enable or motivate improved performance on the shop floor, and track the performance of their value streams. Aero Gear's critical operational concerns were meeting delivery schedules, improving inventory turns and achieving production flow. Dave Heintz, Bohica Line Supervisor, talked about the information he used to manage the flow line.

"At the beginning of each week I download the data on required delivery dates and production lead times for Bohica jobs from the Master Production Report to a spreadsheet. I identify the jobs that should be in process that week to achieve delivery targets. I

count the total units in all those jobs and divide by five [days in the week] to get a takt time in pieces per day.[2] *I do separate takt time computations for the soft and hard ends of the line. If a job is currently in the soft end of the line but it really needs to be in the hard end to meet the delivery date, the job is included in both the soft and hard end takt time calculations.*

If the calculated takt time is too high, I'll increase the denominator until I get to an achievable number. For example, if I need to use a denominator of eight to get an achievable takt time, that tells me the line is three days behind schedule."

The target takt time (pieces per day and per week) versus actual performance were posted on display boards on the line daily. Actual shipments versus the monthly goal were also posted. Boards by each machine listed the day's production schedule for the machine.

Takt time was not calculated for the Blue Line. Mike Sullivan, the Blue Line Supervisor, explained.

"I don't compute a takt time because the operations required for each job are so different that pieces aren't equivalent. Instead I use the data on production lead times, shipping target dates, and the current stage of completion for Blue Line jobs from the Master Production Report to create a priority list for each machine on the line."

The job priorities were posted daily on display boards at each machine. Actual shipments compared to the monthly goal were also posted on the line.

Ray Hutt, the Gearbox Assembly Line manager, also used the required delivery date and lead-time to initiate assembly of a gearbox. Monthly shipments versus target, the production schedule, and information on parts shortages were posted on boards in the Assembly area. Ray Hutt talked about managing the Assembly Line.

"The critical factor in Assembly is making sure we have the parts we need, both those we produce and those we buy outside, at the time we need to initiate assembly of a gearbox. So I spend a lot of time monitoring inventory, parts purchases and incoming shipments, and our production of the parts we use in Assembly."

None of the line managers found the Budgeted Hours Report or the Department Performance Report very helpful in managing their lines. Dave Heintz discussed the reports.

"I get the reports and I'll take a look at them, but I don't put much stock in either one. If an employee's smart he can adjust his reporting so his efficiency percentage is never below 100 percent. They never alert me to a problem I don't already know about. If I see an employee has really high efficiency I'll make a point to single them out for praise, but that's about it."

To the extent he used the reports at all, Mike Sullivan focused on the negative outliers.

"If I see that someone has consistently low efficiency percentages, or they're reporting very high proportions of indirect time I'll talk to them about it, but otherwise the reports don't really help me in managing the line."

Aero Gear also posted charts showing a history of monthly performance against monthly targets on Sales, Sales per Employee, and Inventory turns for the value streams, the job shop, and the overall company. Examples of the charts are provided in Exhibit 5. Aero Gear's management received monthly reports on profitability for each flow line and the job shop, but they did not display charts on profitability. The managers were concerned that the cost allocations to the jobs and consequently to the lines or areas producing the jobs, were inadequate.

PRODUCT COSTING

Material costs and any outside processing or outside tooling were directly assigned to the specific job runs requiring the resources. Aero Gear used a simple, labor-based system to allocate conversion costs to jobs. Jobs were charged a shop-wide standard rate of $75 per direct labor hour that was applied based on actual direct labor hours.[3]

ALTERNATIVES BEING CONSIDERED

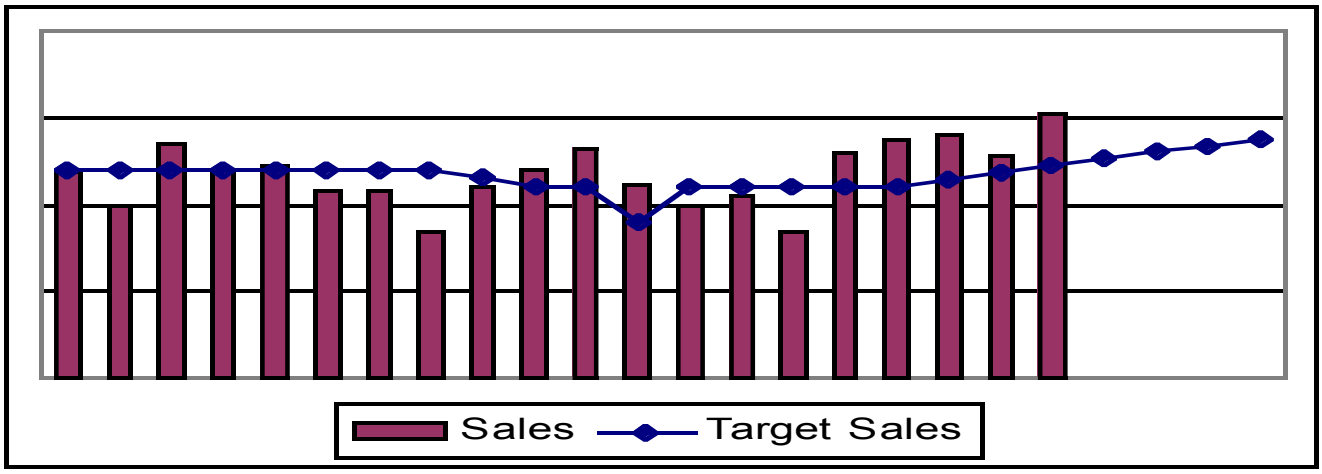
Through the spring and summer of 2001, Aero Gear's managers considered possible changes of their cost allocation methods and their performance measurement system. One suggestion was to adopt throughput accounting. Inventory would be recorded at the cost of materials and outside processing and all conversion costs would be treated as period costs. If throughput accounting were adopted, inventory would have to be adjusted in aggregate to an absorption-cost value for financial and tax reporting. Aero Gear's managers were uncomfortable with throughput accounting. As John McDermott noted, "I don't see how this system will let us assess the profitability of individual jobs."

[2] Takt time is a measure of the rate of production needed to satisfy customer demand. Takt time is used in Lean production environments to set the rate of production to avoid overproduction and minimize inventory while meeting customer demand.

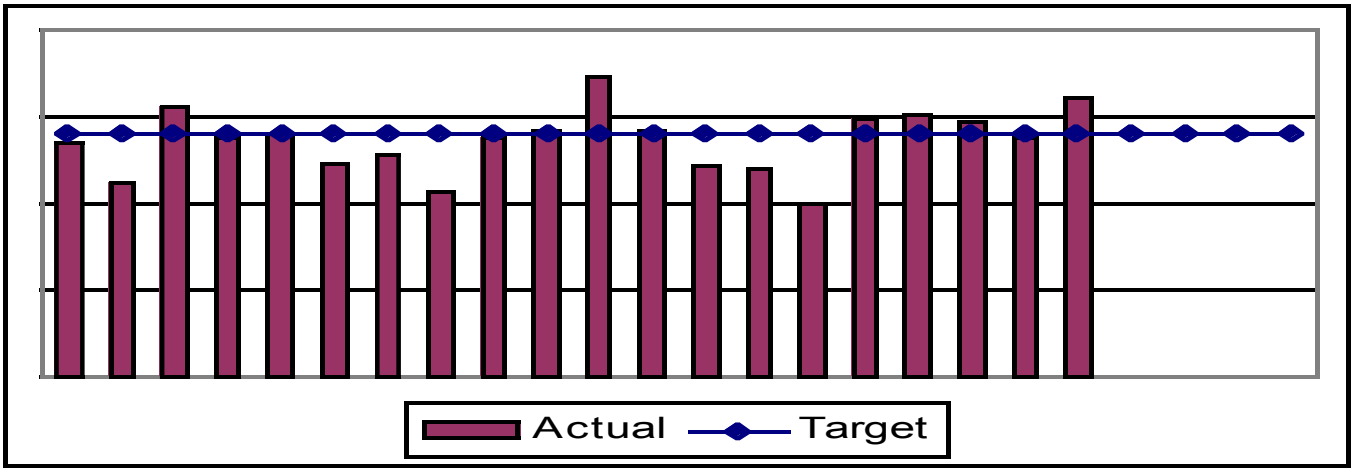
[3] Direct labor hours included time devoted to machine setups and making tooling as well as time performing and monitoring machining operations that make the products.

Exhibit 5 - Aero Gear's Performance Measurement Charts*

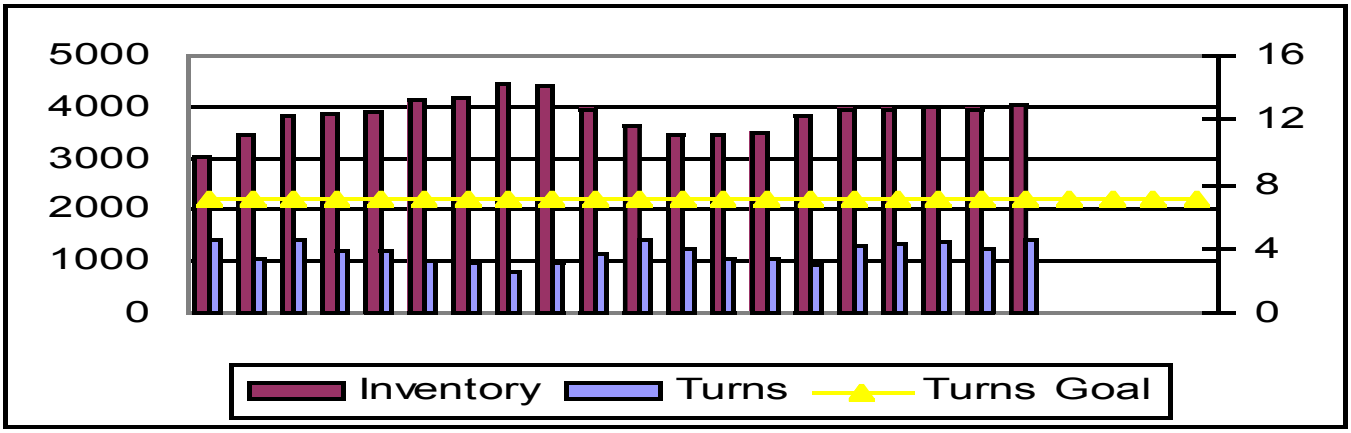
Sales



Total Sales per Employee



Inventory Turns



* Data are for the months January 2000 – August 2001. Y axis labels have been removed from the Sales and Total Sales per Employee Charts, and altered on the Inventory Turns Chart at the request of Aero Gear's management.

Activity-based costing was another possible approach. Activity-based costing would require Aero Gear to identify activities, allocate costs to the activities, and identify cost drivers for each activity. Aero Gear had been identifying and allocating costs to the value streams and the job shop, but it was currently using a single overhead pool for allocating costs to products.

Aero Gear's management heard that some lean companies were using cycle (or throughput time) to allocate costs to products. Cycle time is the entire time required to complete a product, including queue times, setup times, processing times, and inspection times. Using cycle time supports lean business practice because it penalizes the waste of work in process sitting in queues. As waste is eliminated, cycle times are reduced and products are rewarded with lower allocated costs.

Aero Gear's managers were concerned that a cycle time system would unfairly penalize parts sent out for plating. John McDermott expressed their reservations.

"Suppose we send a job out for plating and it's out of the shop for two weeks. It's going to be picking up our shop overhead costs for all that time even though it's not in the shop at all. With that cost on top of the plating cost, How will we ever show a profit on a plated part? If we price to recover those costs, we may lose all our business for parts requiring plating."

Restricting the cycle time measure to time in the facility or time in flow lines was offered as an alternative in response to the reservations about jobs being charged overhead for time outside the facility. The latter variation would require Aero Gear to develop cost pools and cycle time rates for each flow line and the job shop and to track the time spent by each job in each area. In 2001, the labor reports were the only measure of time associated with the jobs.

Mark Tustin, an engineer, conducted a one-month study of the actual times jobs spent in the Bohica and Blue Lines. He found that queue time was a considerable proportion of the time for several jobs. Aero Gear's managers were concerned that costs of individual jobs could be distorted if queue times were included in the allocation base because queue time for one job could be caused by run problems on another job. Doug Rose expressed his concern.

"Why should one job pay a cost penalty for problems caused by another job? I think for where we stand right now, individual job costs assigned using a cycle time would be too arbitrary. I think we have to do a better job of achieving flow before cycle time will work for us."

To avoid the anticipated problem with individual job cost measurement, Aero Gear's managers considered dropping queue time from the cycle time measure. This would leave tool hours, setup hours and processing hours.

Using cycle time or any of its components would require creating a system to track the time jobs spent each area. Moving away from total cycle time actually made the tracking more complex. For example, time in line requires one entry for the time the job enters the line and another when the job exits. If setup time and processing time are tracked, each requires two entries for every operation performed on the job while it is on the line. Still, tracking setup times and processing time for each job would involve significantly fewer entries than the current labor reporting system if detailed labor reporting were eliminated. For most employees on most days, all their time would be on a single line. John McDermott expressed concern with losing detailed information about how employees were spending their time.

"If we no longer have employees account for how they're spending their time, how can we be sure they're spending their time on processing instead of wasting time on other activities? How will we be able to tell if we're becoming more efficient?"

Doug Rose elaborated.

"We believe our employees are hard working and they want to do a good job, but I think having that time measure and the accountability adds a little more pressure and gets them to focus a bit more and push a little harder."

The tracking could be avoided if Aero Gear used the budgeted times that Everett Smith was developing as standard times. If standard costs were used in the absence of actual tracking, Aero Gear would have to create a process to maintain and update standards. They would also still need a means of identifying percentage of completion for work in process at the end of a period. It was also useful to be able to track progress of jobs on an ongoing basis to be able to respond to customer inquiries, though tracking job progress for customer inquiries would not require cost allocations.

MOVING FORWARD

As Aero Gear approached the close of its fiscal year in the fall of 2001, Doug, John and Everett considered their product costing system and the performance measures they were posting on the shop floor. Should the product costing system be changed? If so, what change should they make? Should they hold off making a change despite their dissatisfaction with the current system and continue to look for a better product costing method?

Were there any additional measures they could collect and use to measure flow line performance? For example, they had begun reporting sales on a weekly basis to avoid month-end spikes and smooth flow. Set-up times were and incidences of unscheduled machine down-time could be found in the detail of their current reports, but the information was not summarized and reported in a useful form, nor was it being posted in the flow lines. Measures such as scrap or defect rates and on-time delivery were also being considered. Managers were concerned that using promise dates for on-time delivery would encourage quoting long lead times. On the other hand, customer requests were sometimes unreasonable, even impossible. Was it fair to punish a flow line if a customer suddenly wanted a job "yesterday" and decided another job should be delayed? Should additional operational or financial measures be gathered, and if so, would posting the measures on the shop floor provide additional motivation for improvement?

ABOUT IMA

With a worldwide network of nearly 60,000 professionals, IMA is the world's leading organization dedicated to empowering accounting and finance professionals to drive business performance. IMA provides a dynamic forum for professionals to advance their careers through Certified Management Accountant (CMA®) certification, research, professional education, networking and advocacy of the highest ethical and professional standards. For more information about IMA, please visit www.imanet.org.